# Sadik Bera Yuksel

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## PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Engineering specializing in machine learning, with a focus on reinforcement learning and autonomous systems. Experienced in developing and deploying learning-based algorithms for robotic decision-making and control in complex environments. Proficient in Python and modern ML frameworks such as PyTorch, with strong experience in building, training, and evaluating machine learning models for real-world applications, as well as managing research and technical projects.

## EDUCATION

**Ph.D. in Computer Engineering** — Expected May 2028
Northeastern University, Boston, MA | GPA: 4.0/4.0

**M.S. in Computer Engineering** — May 2025
Northeastern University, Boston, MA | GPA: 4.0/4.0
**Concentration:** Computer Vision, Machine Learning, and Algorithms

**Bilkent University** — June 2023
B.S. in Electrical and Electronics Engineering

**Korea University** — February - June 2022
Study Abroad Program, Electrical and Electronics Engineering

## PROFESSIONAL EXPERIENCE

**Northeastern University** — Boston, MA
*Doctoral Researcher (Advisor: Prof. Derya Aksaray)* — *September 2023 - Present*

- Conducted research on safe autonomy by integrating Signal Temporal Logic into reinforcement learning and robotic foundation models for constrained decision-making under spatio-temporal tasks.
- Trained and evaluated deep reinforcement learning policies using PyTorch and adapted them within a foundation-model-based framework for robust action selection under temporal logic specifications.
- Developed a temporal-logic-guided motion planning framework with contingency constraints to ensure robust autonomous navigation under uncertainty with probabilistic guarantees.
- Deployed the planning algorithm on a real drone in a GPS-denied environment, integrating perception, state estimation, control, and runtime monitoring for autonomous task execution.

**Northeastern University** — Boston, MA
*Teaching Assistant* — *January - May 2026*

- Teaching Assistant for EECE5644: Introduction to Machine Learning and Pattern Recognition, assisting with instruction, grading, and student support for machine learning theory and applications.

**ASELSAN** — Ankara, Turkey
*Transmitter and Receiver Design Intern* — *June - August 2022*

- Designed and manufactured an RF diplexer using Keysight Genesys and PCB layout tools.
- Validated diplexer performance through S-parameter analysis with a vector network analyzer.

**HAVELSAN** — Ankara, Turkey
*Robotics and Autonomous Systems Intern* — *June - August 2021*

- Implemented autonomous navigation for a differential drive robot in a simulation environment using ROS and Python.
- Developed a vision-based face recognition system on Qualcomm Robotics RB5 platform using OpenCV.

## PUBLICATIONS

- **S.B. Yüksel** and D. Aksaray, “Specification-Aware Distribution Shaping for Robotics Foundation Models”, under review at *International Conference on Intelligent Robots and Systems (IROS)*, 2026. arXiv:2603.17969
- **S.B. Yüksel**, A.T. Buyukkocak and D. Aksaray, “Shielded Reinforcement Learning Under Dynamic Temporal Logic Constraints”, to appear in *American Control Conference (ACC)*, 2026.
- X. Lin, **S.B. Yüksel**, Y. Yazıcıoğlu, and D. Aksaray, “Probabilistic Satisfaction of Temporal Logic Constraints in Reinforcement Learning via Adaptive Policy-Switching”, Proc. of the Learning for Dynamics & Control (L4DC), 2025.
- **S.B. Yüksel**, A. Taheri, Y. Yazıcıoğlu, and D. Aksaray, “Robust Temporal Logic Planning Under Contingency Constraints”, International Conference on Automation Science and Engineering (CASE), 2025.

## SKILLS

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|---|---|
| **Programming Languages** | Python, C++, MATLAB, Assembly |
| **Robotics & Control** | Reinforcement Learning, Motion Planning, State Estimation, Control Systems, SLAM, Sensor Fusion, MuJoCo, ROS |
| **Machine Learning & AI** | PyTorch, JAX, CUDA, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV |
| **Tools and Software** | Linux, Git, Docker, LTSpice, Vivado, Diptrace |

## ACADEMIC PROJECTS

**Sketch-to-Image Translation for Human Identification** Spring 2025
*Northeastern University*

- Designed and trained a Generative Adversarial Network (GAN) model in PyTorch for image generation from sketches, yielding a 28.2% increase in cosine similarity and a 3.5% improvement in SSIM over the baseline, which indicates enhanced feature-level and structural similarity to ground truth images.

**Real-Time Localization and Mapping for Autonomous Vehicles** Fall 2023
*Northeastern University*

- Developed and deployed a real-time visual-inertial SLAM pipeline using ORB-SLAM3 in ROS, fusing camera and IMU measurements for robust localization and mapping in driving environments using real vehicle data.

**Indoors Drone Localization Using UWB Sensors** Spring 2023
*Bilkent University*

- Developed an indoor drone localization system in Python that fuses Ultra-Wideband (UWB) distance data with IMU measurements using trilateration and filtering algorithms to determine accurate real-time position estimates, achieving an average localization error of 0.26 meters in real time.
- Deployed the algorithm on the drone’s onboard computer and created a real-time monitoring interface for ground-based visualization with an update rate of 20 Hz.

**Vision-Language Model for Image Understanding** Spring 2023
*Bilkent University*

- Trained LSTM and GRU-based models with attention mechanisms using PyTorch to generate image captions using the Flickr8k and Flickr30k datasets, achieving a BLEU score of 0.5168
- Optimized model performance through hyperparameter tuning with the Optuna library in Python and compared Greedy Search and Beam Search caption generation methods to evaluate caption quality.